

Manual handtrace (Exercise Q-1)

$$A = [1, 2, 3, 4]$$

$$B = [6, 2, 1, 9]$$

$$N = 2$$

$$\text{TILEWIDTH} = 2$$

row col

$$A[0][0]$$

$$A[0][1]$$

$$A[1][0]$$

$$A[1][1]$$

flat index (row major)
(row * N + col)

$$0(2) + 0 = 0 \rightarrow A[0] = 1$$

$$0(2) + 1 = 1 \rightarrow A[1] = 2$$

$$1(2) + 0 = 2 \rightarrow A[2] = 3$$

$$1(2) + 1 = 3 \rightarrow A[3] = 4$$

row col

$$B[0][0]$$

$$B[0][1]$$

$$B[1][0]$$

$$B[1][1]$$

flat index (col major)
(col * N + row)

$$0(2) + 0 = 0 \rightarrow B[0] = 6$$

$$1(2) + 0 = 2 \rightarrow B[2] = 1$$

$$0(2) + 1 = 1 \rightarrow B[1] = 2$$

$$1(2) + 1 = 3 \rightarrow B[3] = 9$$

Now, lets go by the code line by line

$$\text{We will have } \text{gridDim} = (N / \text{TILEWIDTH}, N / \text{TILEWIDTH}) \\ = (1, 1)$$

formula:

$$\text{row} = b_y * \text{TW} + t_y \quad b_x = 0, b_y = 0 \quad (\text{only one block})$$

$$\text{col} = b_x * \text{TW} + t_x$$

Thread (tx=0, ty=0)

1. row = $0 * 2 + 0 = 0$

2. col = $0 * 2 + 0 = 0$

3. Mds load:

address into A = $[0][0]$

value fetched = 1

stored into Mds[0][0]

4. Nds load (corner-turned):

address into B = $[0][0]$

value fetched = 6

stored into Nds[0][0]

- Sync threads();

5. Compute loop (k = 0, then k = 1):

k=0: Mds[ty][0] * Nds[0][tx] = * =

k=1: Mds[ty][1] * Nds[1][tx] = * =

Pvalue = sum of both =

6. Final write:

P[row*N + col] = Pvalue =

Thread (tx=0, ty=1)

1. row = $0 * 2 + 1 = 1$

2. col = $0 * 2 + 0 = 0$

3. Mds load:

address into A = $[1][0]$

value fetched = 3

stored into Mds[1][0]

4. Nds load (corner-turned):

address into B = $[1][0]$

value fetched = 2

stored into Nds[1][0]

- Sync threads();

5. Compute loop (k = 0, then k = 1):

k=0: Mds[ty][0] * Nds[0][tx] = * =

k=1: Mds[ty][1] * Nds[1][tx] = * =

Pvalue = sum of both =

6. Final write:

P[row*N + col] = Pvalue =

All these threads
are parallel

Thread (tx=1, ty=0)

1. row = $0 * 2 + 0 = 0$

2. col = $0 * 2 + 1 = 1$

3. Mds load:

address into A = $[0][1]$

value fetched = 2

stored into Mds[0][1]

4. Nds load (corner-turned):

address into B = $[0][1]$

value fetched = 1

stored into Nds[0][1]

- Sync threads();

5. Compute loop (k = 0, then k = 1):

k=0: Mds[ty][0] * Nds[0][tx] = * =

k=1: Mds[ty][1] * Nds[1][tx] = * =

Pvalue = sum of both =

6. Final write:

P[row*N + col] = Pvalue =

Thread (tx=1, ty=1)

1. row = $0 * 2 + 1 = 1$

2. col = $0 * 2 + 1 = 1$

3. Mds load:

address into A = $[1][1]$

value fetched = 4

stored into Mds[1][1]

4. Nds load (corner-turned):

address into B = $[1][1]$

value fetched = 9

stored into Nds[1][1]

- Sync threads();

5. Compute loop (k = 0, then k = 1):

k=0: Mds[ty][0] * Nds[0][tx] = * =

k=1: Mds[ty][1] * Nds[1][tx] = * =

Pvalue = sum of both =

6. Final write:

P[row*N + col] = Pvalue =

Now we have the entire Mds and Nds tiles filled. Now we can run the 5th step

$$Mds[0][0] = 1$$

$$Mds[0][1] = 2$$

$$Mds[1][0] = 3$$

$$Mds[1][1] = 4$$

$$Nds[0][0] = 6$$

$$Nds[0][1] = 1$$

$$Nds[1][0] = 2$$

$$Nds[1][1] = 9$$

} visual tiles
→

Mds

1	2
3	4

Nds

6	1
2	9

Now let's compute the loop's values

Thread (tx=0, ty=0)

1. row = $0 * 2 + 0 = 0$
2. col = $0 * 2 + 0 = 0$

3. Mds load:

address into A = $[0][0]$

value fetched = 1

stored into Mds[0][0]

4. Nds load (corner-turned):

address into B = $[0][0]$

value fetched = 6

stored into Nds[0][0]

- sync threads();

5. Compute loop (k = 0, then k = 1):

k=0: $Mds[ty][0] * Nds[0][tx] = 1 * 6 = 6$

k=1: $Mds[ty][1] * Nds[1][tx] = 2 * 2 = 4$

Pvalue = sum of both = $6 + 4 = 10$

- sync threads()

6. Final write:

$P[row*N + col] = Pvalue = P[0] = 10$

Thread (tx=0, ty=1)

1. row = $0 * 2 + 1 = 1$
2. col = $0 * 2 + 0 = 0$

3. Mds load:

address into A = $[1][0]$

value fetched = 3

stored into Mds[1][0]

4. Nds load (corner-turned):

address into B = $[1][0]$

value fetched = 2

stored into Nds[1][0]

- sync threads()

5. Compute loop (k = 0, then k = 1):

k=0: $Mds[ty][0] * Nds[0][tx] = 3 * 6 = 18$

k=1: $Mds[ty][1] * Nds[1][tx] = 4 * 2 = 8$

Pvalue = sum of both = $18 + 8 = 26$

- sync threads()

6. Final write:

$P[row*N + col] = Pvalue = P[2] = 26$

All these threads
are parallel

Thread (tx=1, ty=0)

1. row = $0 * 2 + 0 = 0$
2. col = $0 * 2 + 1 = 1$

3. Mds load:

address into A = $[0][1]$

value fetched = 2

stored into Mds[0][1]

4. Nds load (corner-turned):

address into B = $[0][1]$

value fetched = 1

stored into Nds[0][1]

- sync threads()

5. Compute loop (k = 0, then k = 1):

k=0: $Mds[ty][0] * Nds[0][tx] = 1 * 1 = 1$

k=1: $Mds[ty][1] * Nds[1][tx] = 2 * 9 = 18$

Pvalue = sum of both = $1 + 18 = 19$

- sync threads()

6. Final write:

$P[row*N + col] = Pvalue = P[1] = 19$

Thread (tx=1, ty=1)

1. row = $0 * 2 + 1 = 1$
2. col = $0 * 2 + 1 = 1$

3. Mds load:

address into A = $[1][1]$

value fetched = 4

stored into Mds[1][1]

4. Nds load (corner-turned):

address into B = $[1][1]$

value fetched = 9

stored into Nds[1][1]

- sync threads()

5. Compute loop (k = 0, then k = 1):

k=0: $Mds[ty][0] * Nds[0][tx] = 3 * 1 = 3$

k=1: $Mds[ty][1] * Nds[1][tx] = 4 * 9 = 36$

Pvalue = sum of both = $3 + 36 = 39$

- sync threads()

6. Final write:

$P[row*N + col] = Pvalue = P[3] = 39$

Therefore, our final P array:

$$P[0] = 10$$

$$P[1] = 19$$

$$P[2] = 26$$

$$P[3] = 39$$

$$\left. \begin{array}{l} P[0] = 10 \\ P[1] = 19 \\ P[2] = 26 \\ P[3] = 39 \end{array} \right\} \rightarrow \begin{bmatrix} 10 & 19 \\ 26 & 39 \end{bmatrix}$$

Now, let's hand compute

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 6 & 1 \\ 2 & 9 \end{bmatrix} = \begin{bmatrix} 10 & 19 \\ 26 & 39 \end{bmatrix}$$

Yes, both Match

